LESSON PLAN

DATE: Feb. 23th, 2024

PREPARE

Objectives: By the end of the lesson the students will be able to…

1. Consolidate their knowledge on linear independence/dependence

2. Understand proof by contrapositive

3. Understand the definition of basis; understand that every subspace of R^n has a basis

Preparation (materials, resources, practice):

- Materials: lecture notes (attached), tablet

- Practice in front of Leslie on Feb. 19th in Gross 330

Personal goals (what I'm working on in my teaching):

- ➢ Pace and tone of my lecturing
- ➢ Live writing on tablet

INSTRUCT

Intro:

Bridge-in to create interest and motivation to learn:

- Introduce basis as (arguably) the most powerful tool in linear algebra: it endows abstract vector spaces/subspaces with coordinates so that one is able to work concretely

State learning objectives and connections to course as a whole:

1. Wrap up "linear independence" from last time
2. Introduction to basis

Review:

1. Definition of linear independence/independence
2. Proposition proven last time: a family of vectors is linearly independent if and only if every vector in the span can be written as a linear combination of these vector uniquely
3. Two examples of linear independence

Learning Activities:

What you will do, what they will do, check for understanding (assess). Connect activities to each other. Repeat.

Activity:

Prove that if a family of k vectors is linearly independent, then adding another vector into the family yields a linearly independent family if and only if the new vector is not in the span of the k vectors.

Connect:
Introduce and/or recall the proof technique: proof by contrapositive

Activity:
Introduce the definition of basis with examples

Connect:
1. *Standard basis in R^n*
2. *Consequences from a nonsingular matrix*

Activity:
Prove that every subspace of R^n other than the zero space has a basis

Connect:
1. *Standard basis in R^n*
2. *Consequences from a nonsingular matrix*

Outro:

Make connections to rest of course or next lesson:

1. We will apply basis to concrete problems shortly;
2. We will prove that for a fixed vector space, the number of elements in any of its bases is the same and use this fact to define the dimension of a vector space formally

When you finish writing your plan, use this checklist to review it:

At the beginning, I plan to:

1. Use a bridge-in to capture students' interest and motivate them to learn ..... ❑
2. Make the learning outcomes clear. ..... ❑
3. Assess prior learning and student expectations ..... ❑

In the middle, I plan to:

1. Use strategies to actively involve students in the learning process ..... ❑
2. Use a variety of media to illustrate concepts and processes ..... ❑
3. Ensure that the lesson flows easily and logically ..... ❑
4. Ensure that students are learning material that is meaningful and new ..... ❑
5. Provide opportunities for practice and feedback ..... ❑
6. Review and build on related material ..... ❑

At the end, I plan to:

Provide the proper closure students find important. To do this, I will:

1. Assess what students have learned ..... ❑
2. Summarize the lesson ..... ❑
3. Connect the lesson to real life and/or the next lesson ..... ❑

Feb. 23th

MATH 221/721.

<u>Goals.</u>

① Wrap up "linear independence"

② Introduction to basis.

<u>Last time:</u>

<u>Defn.</u> A family of vectors $\{\vec{v}_1 \cdots \vec{v}_k\}$ is called <u>linear independent</u> if:

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = 0 \implies c_1 = c_2 = \cdots = c_k = 0$$

By contrast, $\{\vec{v}_1, \cdots, \vec{v}_k\}$ is called <u>linearly dependent</u> if it is NOT lin. indep.

i.e. there exist $c_1, c_2, \cdots c_k$ not all equal to 0 such that $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = 0$

<u>Prop.</u> $\{\vec{v}_1 \cdots \vec{v}_k\}$ linearly independent $\iff$

when $\vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$ for some $c_1, c_2 \cdots, c_k$, then

i.e. $v \in \mathrm{Span}\{v_1 \cdots v_k\}$

$c_1, c_2 \cdots, c_k$ are unique.

<u>Remark:</u> In other words,

$\{\vec{v}_1 \cdots \vec{v}_k\}$ linearly independent $\iff$

coordinates

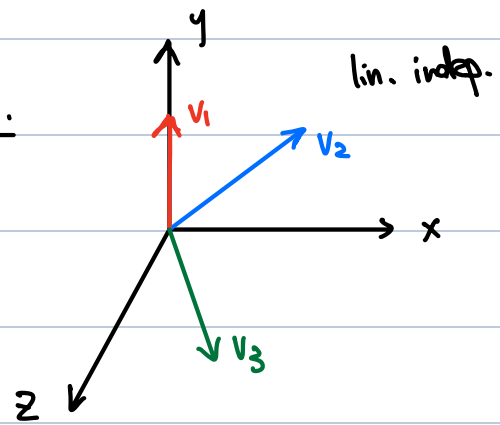
$$\mathrm{Span}(v_1 \cdots v_k) \xrightarrow{\text{bijection}} \left\{ \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} \,\middle|\, c_1, c_2 \cdots c_k \in \mathbb{R} \right\} = \mathbb{R}^k.$$

i.e., $\{v_1 \cdots v_k\}$ linearly independent iff $\mathrm{Span}(v_1 \cdots v_k)$ <u>looks like</u> a piece of $\mathbb{R}^k$ in $\mathbb{R}^n$

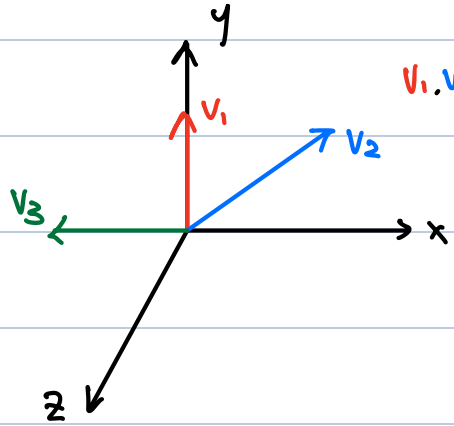
(more details later)

Take this as an intuition, and prove formally using defn/prop.

e.x.



lin. indep.



v_1, v_2, v_3 lin. dep.

BUT, v_1, v_2 lin. indep.

Prop. Suppose $\{v_1, \dots v_k\}$ is lin. indep. and $v \in \mathbb{R}^n$. Then

$\{v, v_1 \cdots v_k\}$ lin. indep. $\iff$ $v \notin \text{Span}\{v_1, \dots, v_k\}$.

Pf. We prove by contrapositive ← If P, then Q $\iff$ If not Q, then not P.

e.g. If $x=2$, then $x^2=4$.

If $x^2 \neq 4$, then $x \neq 2$.

Want to prove: $\{v, v_1 \cdots v_k\}$ lin. dep. $\iff$ $v \in \text{Span}\{v_1, \cdots, v_k\}$

($\Rightarrow$) Given $cv + c_1v_1 + \cdots c_kv_k = 0$ for $c, c_1, \cdots, c_k$ not all equal to 0.

$\Rightarrow$ $c \neq 0$ Why? because otherwise, $c=0$ and

$c_1v_1 + \cdots + c_kv_k = 0$, $c_1, \cdots c_k$ not all 0

$\Rightarrow\Leftarrow$ $\{v_1, \cdots, v_k\}$ lin. indep.

$$cv = -c_1v_1 + \cdots + (-c_kv_k)$$

$$\Rightarrow \quad v = -\frac{c_1}{c}v_1 + \cdots + \left(-\frac{c_k}{c}\right)v_k.$$

$$\Rightarrow \quad v \in \text{Span}\{v_1, \cdots, v_k\}.$$

($\Leftarrow$). $v = c_1v_1 + c_2v_2 + \cdots + c_kv_k$.

$\Rightarrow$ $c_1v_1 + c_2v_2 + \cdots + c_kv_k - v = 0$, where not all coefficients are 0 $(-1 \neq 0)$

$\Rightarrow$ $\{v_1, v_2, \cdots, v_k, v\}$ lin. dep.

Basis. (Arguably the most powerful tool in linear algebra:

Endow abstract vector spaces/subspaces with coordinates to work concretely

Def. Let $V \subset \mathbb{R}^n$ be a subspace. The set of vectors $\{v_1, \cdots, v_k\}$ is called a <u>basis</u> of V if:

(i) $v_1 \cdots v_k$ span V, i.e. $V = \mathrm{Span}(v_1, \cdots, v_k)$.

(ii) $\{v_1, \cdots, v_k\}$ lin. indep.

Ex. ① <u>Standard basis of $\mathbb{R}^n$</u>: $e_1 = (1, 0, \cdots, 0)$ $\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

$e_2 = (0, 1, 0 \cdots, 0)$ $\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$

$\vdots$

$e_n = (0, 0 \cdots 0, 1)$ $\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

② $\{v_1, \cdots, v_k\}$ lin. indep $\Rightarrow$ $v_1, \cdots v_k$ basis of $\mathrm{Span}(v_1, \cdots, v_k)$

③ <u>nonsingular</u> $A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}$ $n \times n$ matrix

$\Leftrightarrow$ $\forall\, b \in \mathbb{R}^n$, $Ax = b$ has a unique solution

$\Leftrightarrow$ Every b can be written as linear combo of $v_1 \cdots v_n$ uniquely

$\Leftrightarrow$ $v_1, \cdots, v_n$ lin. indep. & $\mathrm{Span}(v_1, \cdots v_n) = \mathbb{R}^n$

$\Leftrightarrow$ $\{v_1, \cdots, v_n\}$ is a basis of $\mathbb{R}^n$.

Do not use augment.

$$\begin{pmatrix}1\\2\\1\end{pmatrix}\begin{pmatrix}1\\1\\2\end{pmatrix}\begin{pmatrix}1\\0\\2\end{pmatrix} \quad \begin{pmatrix}1&1&1\\2&1&0\\1&2&2\end{pmatrix} \overset{\text{RREF}}{\rightsquigarrow} \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}. \Rightarrow \text{basis of } \mathbb{R}^3.$$

$$\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}1\\1\\0\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}. \quad \begin{pmatrix}1&1&1\\0&1&0\\1&0&1\end{pmatrix} \overset{\text{RREF}}{\rightsquigarrow} \begin{pmatrix}1&0&1\\0&1&0\\0&0&0\end{pmatrix}. \Rightarrow \underline{\text{not}} \text{ basis of } \mathbb{R}^3.$$

<u>Thm</u>: Every subspace $V \subset \mathbb{R}^n$ <u>other than $\{0\}$</u> has a basis.

<u>Pf</u>: Since $V \neq \{0\}$, we can pick a vector $v_1 \overset{\neq 0}{\in} V$.

If $V = \text{span}\{v_1\}$, pf is complete.

If not, pick $v_2 \in V \setminus \text{span}(v_1)$.

$A \setminus B$: means the complement of B in A, i.e. $\{x \in A \mid x \notin B\}$

If $V = \text{Span}(v_1, v_2)$, pf is complete.

Otherwise, pick $v_3 \in V \setminus \text{span}(v_1, v_2)$.

$\vdots$

Do so iteratively...

The process either ends before we have chosen $v_1, \cdots, v_n$, i.e. we have chosen $v_1, \cdots v_k$, $k < n$ and the pf complete.

<u>OR</u> We have chosen $v_1, \cdots, v_n$.

If latter, we claim that $V = \text{Span}(v_1, \cdots, v_n)$.

<u>Why</u>? If not, then there exists $v \notin \text{Span}(v_1, \cdots v_n)$

By Prop above, $\{v, v_1 \cdots v_n\}$ lin. indep in $\mathbb{R}^n$.

i.e. If $A = \begin{pmatrix} | & | & & | & | \\ v_1 & v_2 & \cdots & v_n & v \\ | & | & & | & | \end{pmatrix}$ $n \times (n+1)$ matrix.

then $Ax = b$ has a unique solution <u>whenever it has a solution</u>.

$\Leftrightarrow \quad \text{rank}(A) = n+1 \qquad (\Rightarrow \Leftarrow)$

$\Rightarrow \quad V = \text{Span}(v_1, \cdots, v_n). \qquad v_1, \cdots v_n$ basis of V. $\square$

<u>Cor.</u> Let $V \subset \mathbb{R}^n$ be a subspace. Then $(V^\perp)^\perp = V$

<u>Pf.</u> Let $\{v_1, \cdots, v_k\}$ be a basis of V. Consider the $n \times k$ matrix

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_k \\ | & & | \end{pmatrix}. \quad \text{Then, } V = R(A)$$

$$\Rightarrow V^\perp = R(A)^\perp = N(A)$$

$$\Rightarrow (V^\perp)^\perp = N(A)^\perp = R(A) = V.$$